COMPACT OBJECTS IN THE DERIVED CATEGORY OF A GROTHENDIECK ABELIAN CATEGORY ARE SUMMANDS OF FINITE COMPLEXES BUILT FROM GENERATORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the proof of the improvement claim in the last paragraph and the splitting of the idempotent; the author's closing clause is retained verbatim.

Editorial note. Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree (Homology and Derived Categories). Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invoking line of the pinned chapter, so no load-bearing mathematics is replaced by a stub.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Compact object of a triangulated category. Same chapter, the definition of a compact object K in D(A) as one for which Hom_{D(A)}(K, -) commutes with direct sums. Supplies the hypothesis and, through the same property, the finite factorisations on which the whole proof rests.
- Canonical truncations tau_{>= n} and stupid truncations sigma_{>= n}. Homology, the canonical and stupid truncation functors on complexes and their basic properties. Supplies the two filtrations of the representing complex, whose finite stages replace K up to factorisation of the identity.
- Bounded above resolution by direct sums of generators and the functoriality of the derived category. Derived Categories, the existence of a bounded above resolution whose terms are direct sums of objects of S and the properties of countable direct sums in the derived category. Supplies the representing complex with terms in direct sums of S and the passage from the union of stupid truncations to a factorisation of the identity.
- Idempotents in a triangulated category and the kernel of an endomorphism. Derived Categories, the splitting of an idempotent e via Ker(e) and Ker(1 - e) and the homological behaviour of represented functors. Supplies the

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final step, which produces the required direct summand of a finite complex built from S.

Source reference. Original lemma 0118: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13957–13957. Navigation pointer invoked at proof line 13957; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05T7: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13966–13966. Navigation pointer invoked at proof line 13966; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0949: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13973–13973. Navigation pointer invoked at proof line 13973; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 094A: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13974–13974. Navigation pointer invoked at proof line 13974; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 05QW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13995–13995. Navigation pointer invoked at proof line 13995; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0152: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14021–14021. Navigation pointer invoked at proof line 14021; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0149: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 14023–14023. Navigation pointer invoked at proof line 14023; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be a Grothendieck abelian category. Let $S \subset \mathrm{Ob}(\mathcal{A})$ be a set of objects such that*

(1) *any object of $\mathcal{A}$ is a quotient of a direct sum of elements of S, and*
(2) *for any $E \in S$ the functor $\mathrm{Hom}_{\mathcal{A}}(E, -)$ commutes with direct sums.*

Then every compact object of $D(\mathcal{A})$ is a direct summand in $D(\mathcal{A})$ of a finite complex of finite direct sums of elements of S.

Proof. Assume $K \in D(\mathcal{A})$ is a compact object. Represent K by a complex $K^\bullet$ and consider the map

$$K^\bullet \longrightarrow \bigoplus\nolimits_{n \geq 0} \tau_{\geq n} K^\bullet$$

where we have used the canonical truncations, see Homology, Section 0118. This makes sense as in each degree the direct sum on the right is finite. By assumption

this map factors through a finite direct sum. We conclude that $K \to \tau_{\geq n}K$ is zero for at least one n, i.e., K is in $D^-(R)$.

We may represent K by a bounded above complex $K^\bullet$ each of whose terms is a direct sum of objects from S, see Derived Categories, Lemma 05T7. Note that we have

$$K^\bullet = \bigcup\nolimits_{n \leq 0} \sigma_{\geq n} K^\bullet$$

where we have used the stupid truncations, see Homology, Section 0118. Hence by Derived Categories, Lemmas 0949 and 094A we see that $1 : K^\bullet \to K^\bullet$ factors through $\sigma_{\geq n}K^\bullet \to K^\bullet$ in $D(R)$. Thus we see that $1 : K^\bullet \to K^\bullet$ factors as

$$K^\bullet \xrightarrow{\varphi} L^\bullet \xrightarrow{\psi} K^\bullet$$

in $D(\mathcal{A})$ for some complex $L^\bullet$ which is bounded and whose terms are direct sums of elements of S. Say L^i is zero for $i \notin [a, b]$. Let c be the largest integer $\leq b + 1$ such that L^i a finite direct sum of elements of S for $i < c$. Claim: if $c < b + 1$, then we can modify $L^\bullet$ to increase c. By induction this claim will show we have a factorization of 1_K as

$$K \xrightarrow{\varphi} L \xrightarrow{\psi} K$$

in $D(\mathcal{A})$ where L can be represented by a finite complex of finite direct sums of elements of S. Note that $e = \varphi \circ \psi \in \mathrm{End}_{D(\mathcal{A})}(L)$ is an idempotent. By Derived Categories, Lemma 05QW we see that $L = \mathrm{Ker}(e) \oplus \mathrm{Ker}(1-e)$. The map $\varphi : K \to L$ induces an isomorphism with $\mathrm{Ker}(1 - e)$ in $D(R)$ and we conclude.

Proof of the claim. Write $L^c = \bigoplus_{\lambda \in \Lambda} E_\lambda$. Since L^{c-1} is a finite direct sum of elements of S we can by assumption (2) find a finite subset $\Lambda' \subset \Lambda$ such that $L^{c-1} \to L^c$ factors through $\bigoplus_{\lambda \in \Lambda'} E_\lambda \subset L^c$. Consider the map of complexes

$$\pi : L^\bullet \longrightarrow (\bigoplus\nolimits_{\lambda \in \Lambda \setminus \Lambda'} E_\lambda)[-i]$$

given by the projection onto the factors corresponding to $\Lambda \setminus \Lambda'$ in degree i. By our assumption on K we see that, after possibly replacing Λ' by a larger finite subset, we may assume that $\pi \circ \varphi = 0$ in $D(\mathcal{A})$. Let $(L')^\bullet \subset L^\bullet$ be the kernel of π. Since π is surjective we get a short exact sequence of complexes, which gives a distinguished triangle in $D(\mathcal{A})$ (see Derived Categories, Lemma 0152). Since $\mathrm{Hom}_{D(\mathcal{A})}(K, -)$ is homological (see Derived Categories, Lemma 0149) and $\pi \circ \varphi = 0$, we can find a morphism $\varphi' : K^\bullet \to (L')^\bullet$ in $D(\mathcal{A})$ whose composition with $(L')^\bullet \to L^\bullet$ gives φ. Setting ψ' equal to the composition of ψ with $(L')^\bullet \to L^\bullet$ we obtain a new factorization. Since $(L')^\bullet$ agrees with $L^\bullet$ except in degree c and since $(L')^c = \bigoplus_{\lambda \in \Lambda'} E_\lambda$ the claim is proved. □